

A positive averaging operator whose inverse does not preserve hyperbolicity

A counterexample to Problem 2 of arXiv:0801.1749

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Status. Candidate counterexample, likely valid, pending human review. An explicit infinite-order constant-coefficient differential operator preserves every strictly positive polynomial and has strictly positive-definite moment matrices, but its inverse sends the hyperbolic polynomial x^6 to a polynomial with nonreal zeros. Thus Problem 2 has a negative answer.

1 Source question

Let $D = d/dx$. The source calls a linear operator on $\mathbb{R}[x]$ a *positivity-preserver* if it maps every polynomial positive on all of $\mathbb{R}$ to another such polynomial, and a *hyperbolicity-preserver* if it maps every real-rooted polynomial to a real-rooted polynomial. Problem 2 on source PDF page 3 asks whether every positivity-preserver of the form

$$U = \alpha_0 + \alpha_1 D + \alpha_2 D^2 + \cdots$$

has a hyperbolicity-preserver as its inverse [1].

$$\begin{pmatrix} 1!\alpha_1 & 2!\alpha_2 & 3!\alpha_3 & \cdots & (l+1)!\alpha_{l+1} & \cdots \\ 2!\alpha_2 & 3!\alpha_3 & 4!\alpha_4 & \cdots & (l+2)!\alpha_{l+2} & \cdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ l!\alpha_l & (l+1)!\alpha_{l+1} & (l+2)!\alpha_{l+2} & \cdots & (2l)!\alpha_{2l} & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{pmatrix}$$

is positive definite (resp., positive semi-definite), i.e. all its principal minors are positive (resp. non-negative).

To illustrate the latter result notice that for the operator (2) above one has $1 = \alpha_0 = \alpha_1 = \alpha_2 = \alpha_3 = \dots$ and $\Delta_l = \Pi_{j=1}^l (j!)^2$, $l = 0, 1, \dots$ where Δ_l is the corresponding $(l+1) \times (l+1)$ principal minor, see [6].

Remark 1.7. The major remaining challenge in this area is to classify *all* positivity-preservers. We finish our introduction with this question.

Problem 1. Find a complete classification of positivity-preservers.

We also state a more concrete and (hopefully) simpler question.

Problem 2. Is it true that any positivity-preserver which is an infinite order linear differential operator with constant coefficients has a hyperbolicity-preserver as its inverse?

2 Proof intuition

Every probability measure with all moments gives a constant-coefficient positivity-preserver by averaging translations:

$$(U_\mu p)(x) = \int_{\mathbb{R}} p(x+t) d\mu(t).$$

The two-point average has symbol $\cosh z$, so its formal inverse has symbol $\operatorname{sech} z$. Already

$$\operatorname{sech}(D)x^6 = x^6 - 15x^4 + 75x^2 - 61$$

is not real-rooted. To rule out any objection that this seed measure has finite support and degenerate high-order moment matrices, add one percent of a Gaussian. Strict positive definiteness is restored, while the inverse image of x^6 retains a pair of nonreal conjugate roots.

3 The counterexample

Let γ be standard Gaussian measure and set

$$\mu = \frac{99}{200}(\delta_{-1} + \delta_1) + \frac{1}{100}\gamma. \quad (1)$$

Define

$$(Up)(x) = \int_{\mathbb{R}} p(x+t) d\mu(t) = \left(\frac{99}{100} \cosh D + \frac{1}{100} e^{D^2/2} \right) p(x). \quad (2)$$

Theorem 1. *The operator $U : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ in (2) is an invertible, infinite-order, constant-coefficient positivity-preserver. Every finite Hankel moment form associated with U is positive definite. Nevertheless U^{-1} is not a hyperbolicity-preserver.*

Proof. The translation identity $e^{tD}p(x) = p(x+t)$ and the Gaussian moment formula give (2). Thus U is a constant-coefficient formal differential operator with infinitely many nonzero coefficients. If $p(x) > 0$ for every $x \in \mathbb{R}$, then every integrand $p(x+t)$ is positive, and hence $(Up)(x) > 0$ for every x . Therefore U preserves positivity.

The moments $m_k = \int t^k d\mu(t)$ determine its Hankel forms. For every nonzero polynomial q ,

$$\sum_{i,j} m_{i+j} q_i q_j = \int_{\mathbb{R}} q(t)^2 d\mu(t) \geq \frac{1}{100} \int_{\mathbb{R}} q(t)^2 d\gamma(t) > 0.$$

The final inequality holds because a nonzero polynomial cannot vanish on a set of full Gaussian measure. Thus all finite moment matrices are strictly positive definite.

The formal symbol of U is

$$M(z) = \frac{99}{100} \cosh z + \frac{1}{100} e^{z^2/2}.$$

Since $M(0) = 1$, it has a unique reciprocal formal power series $N(z)$, and $V = N(D)$ is the two-sided inverse of U on $\mathbb{R}[x]$: all series truncate on each input polynomial. Through degree six,

$$\begin{aligned} M(z) &= 1 + \frac{1}{2}z^2 + \frac{17}{400}z^4 + \frac{19}{12000}z^6 + O(z^8), \\ N(z) &= 1 - \frac{1}{2}z^2 + \frac{83}{400}z^4 - \frac{1009}{12000}z^6 + O(z^8). \end{aligned}$$

Consequently

$$V(x^6) = x^6 - 15x^4 + \frac{747}{10}x^2 - \frac{3027}{50}. \quad (3)$$

The input x^6 is hyperbolic. To inspect the output, put $y = x^2$ and multiply the resulting cubic by 50:

$$R(y) = 50y^3 - 750y^2 + 3735y - 3027.$$

Its discriminant is

$$\operatorname{Disc}(R) = -668917845000 < 0. \quad (4)$$

A real cubic with negative discriminant has one real zero and a nonreal conjugate pair. Hence the polynomial in (3) has nonreal zeros and is not hyperbolic. Thus $U^{-1} = V$ is not a hyperbolicity-preserver. $\square$

4 Deep upgrade, novelty, and scope

The first attempt used the transparent seed $U_0 = \cosh D = (e^D + e^{-D})/2$. Its inverse already fails on x^6 ; writing $y = x^2$ factors the corresponding cubic as

$$y^3 - 15y^2 + 75y - 61 = (y - 1)(y^2 - 14y + 61),$$

whose quadratic factor has negative discriminant. The deep upgrade in Theorem 1 mixes in a Gaussian. This forces $\int q^2 d\mu > 0$ for every nonzero q , so the counterexample lies in the strict interior of the moment cone rather than on a finite-support boundary. The symbolic coefficient and discriminant calculations are independently reproduced in `code/verify_counterexample.py`.

A bounded primary-source search used the exact wording of Problem 2, “positivity-preserver”, “hyperbolicity-preserver”, inverse, moment measure, $\cosh(D)$, and $\operatorname{sech}(D)$. It found the source and later moment/generator work, including arXiv:2308.10455 [2], but no primary paper stating this counterexample. Novelty confidence is moderate; correctness confidence is high.

This disproves Problem 2 only. It does not classify all positivity preservers as requested in the much broader Problem 1. The argument uses the source’s algebraic convention for infinite-order operators: formal power series in D act unambiguously on polynomials because only finitely many derivatives survive.

References

- [1] J. Borcea, A. Guterman, and B. Shapiro, *Preserving positive polynomials and beyond*, arXiv:0801.1749.
- [2] P. J. di Dio, *On positivity preservers with constant coefficients and their generators*, arXiv:2308.10455.